

The Future of Embodied AI: Humanoid Robotics

Circulus Robotics, Intel and Red Hat

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Session Abstract

Humanoid robots are transitioning from research labs to the industrial frontline, but this move requires a radical rethink of the compute stack. To operate safely alongside humans, these machines need massive AI throughput, millisecond-level latency, and a bulletproof operating system—all within a power-constrained, "on-robot" footprint.

In this session, we showcase a groundbreaking collaboration between **Intel, Circulus Robotics, and Red Hat**. We will explore how the **Intel Panther Lake** platform (Core Ultra Series 3) provides the 180+ TOPS of NPU and GPU power necessary for real-time "Physical AI." You'll see how **Red Hat Enterprise Linux (RHEL)** provides the real-time deterministic foundation and containerized orchestration required to manage complex **ROS2** workloads.

Join us to see a live look at how this integrated stack bridges the gap between digital intelligence and physical action, enabling the next generation of humanoid workers.



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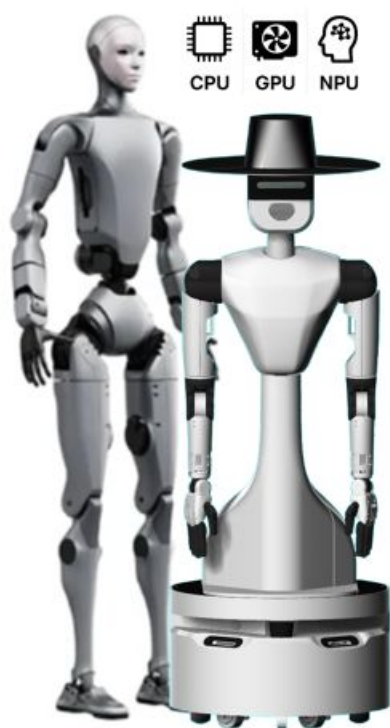
Physical AI Brain on Core Ultra 3



On-device Physical AI for Heterogeneous Robots

Establishing the 'WIntel' Standard: Unified CPU·GPU·NPU Architecture for All Robots

On-device Physical AI for Heterogeneous Robots



CPU GPU NPU

WIntel of Physical AI



Real-Time

+



On-device AI



Task Module

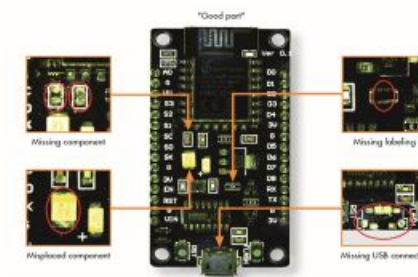


Digital Twin Sync for Dark Factories

Proven Killer Apps for Factory Operations



Pick & Place



Inspection



Worker Safety (PPE)

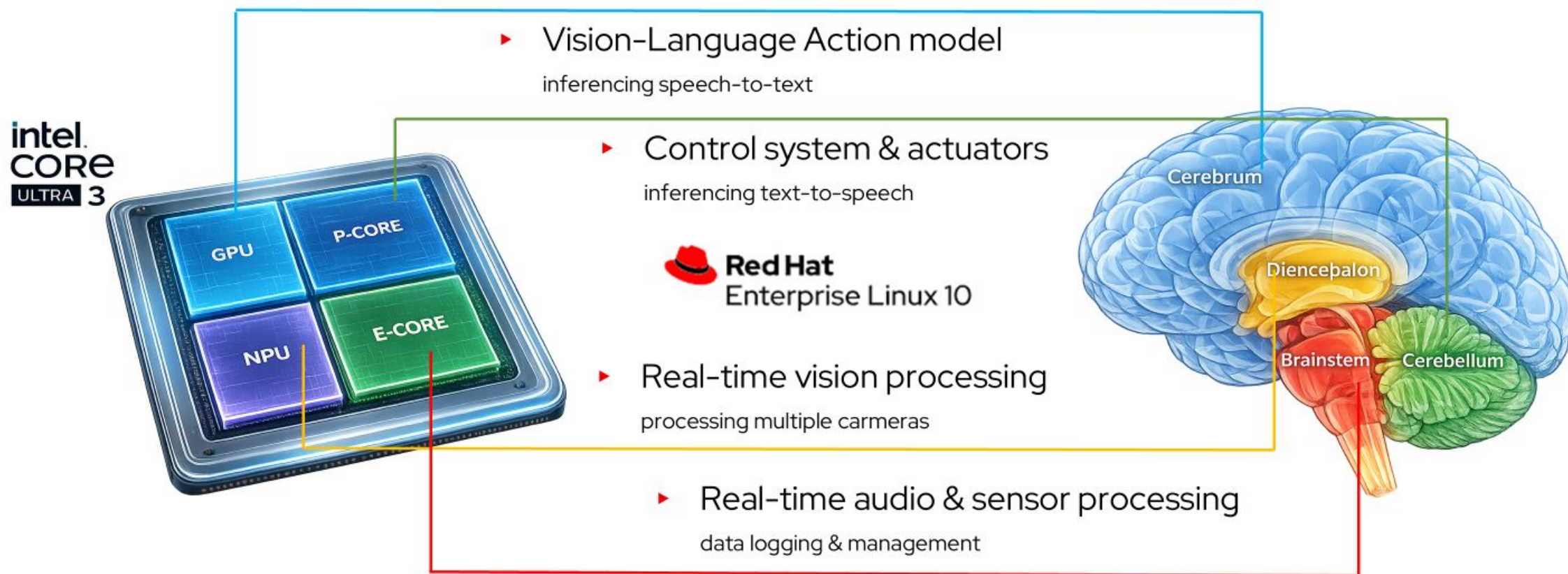


Inventory Check (RFID)



Intel Core Ultra: The Most Human-like Brain for Humanoids

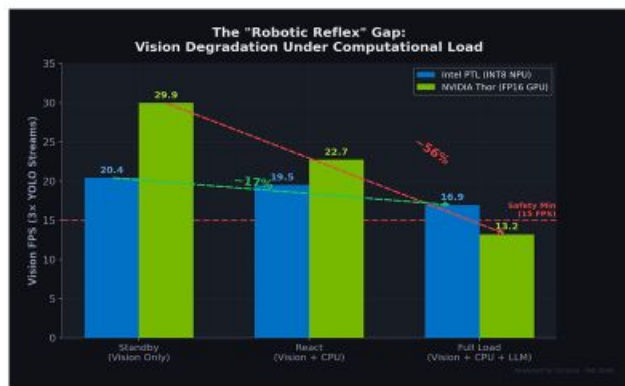
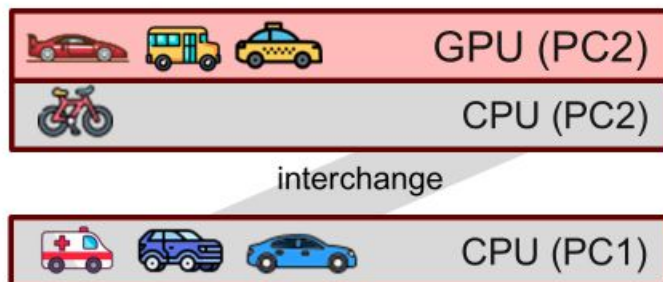
Based on the stability and reliability of RedHat RHEL 10.2



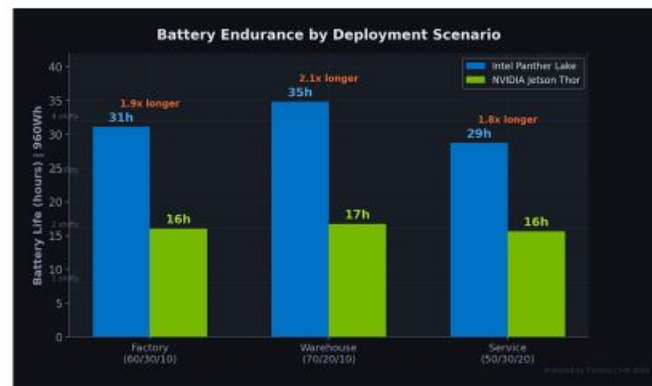
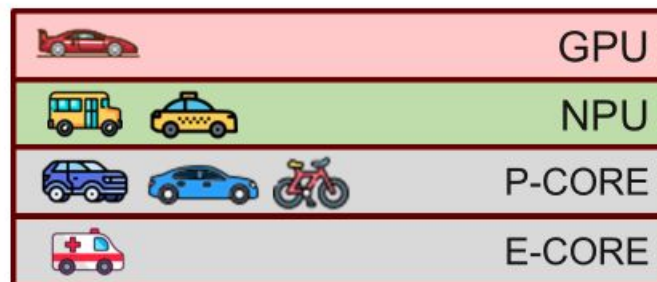
A giant leap toward a human-like brain for humanoid

Beyond Single-Channel: Intel Core Ultra's Multi-Engine Synergy (GPU, NPU, P/E-core)

(AS-IS) NVIDIA Jetson (on Unitree G1)



(TO-BE) Intel Core Ultra 3 (on MCR-1)

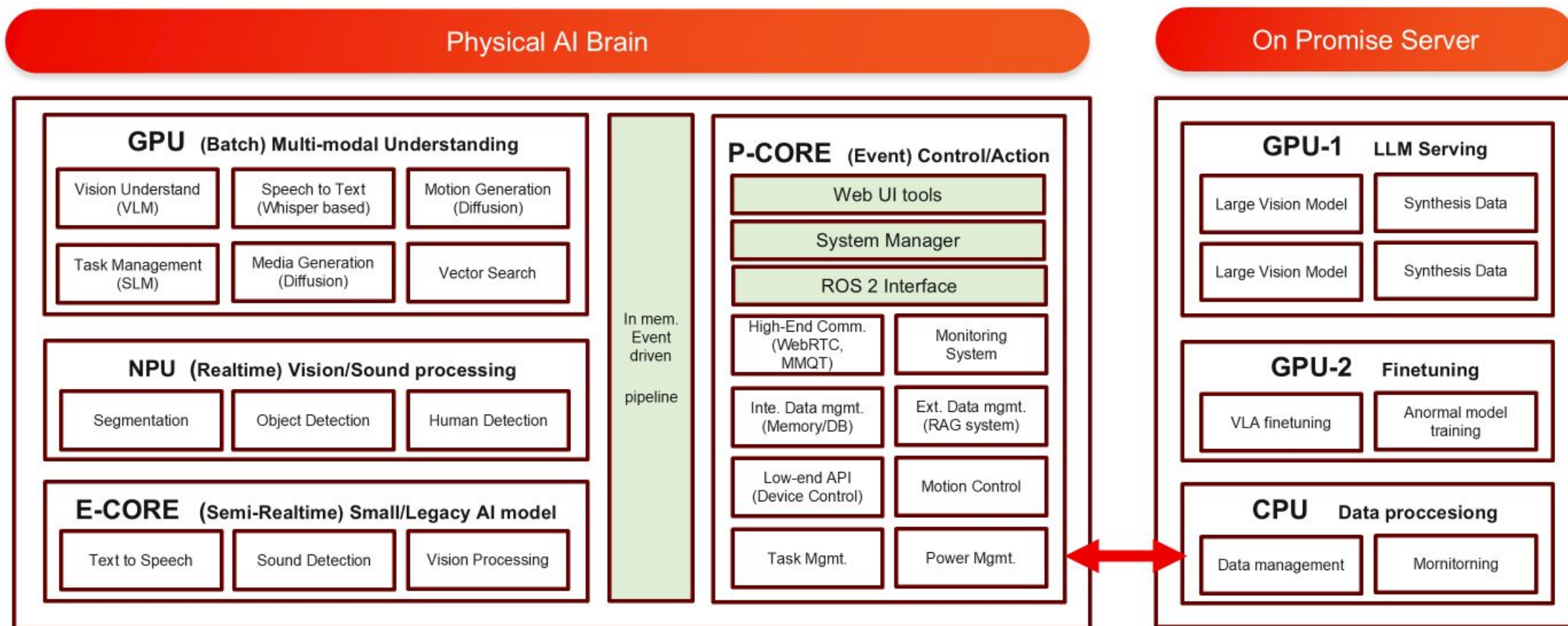


Hybrid Server (B70)



Specialized Engines, One Unified Intelligence

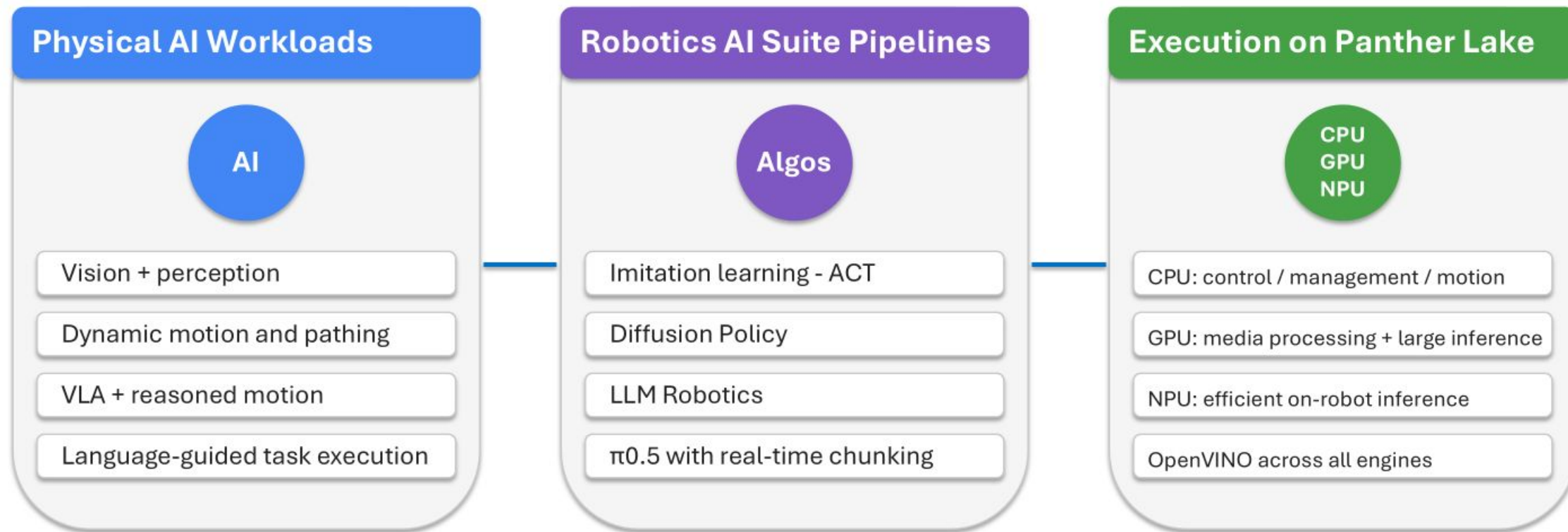
Optimized Orchestration: Distributing Perception, Understanding, and Action



Intel Robotics AI Suite for Humanoids

From humanoid workloads to software pipelines to Panther Lake execution

- Humanoid goals:
- Predict the scene
 - Reason over tasks
 - Learn from demonstrations
 - Act safely in real time

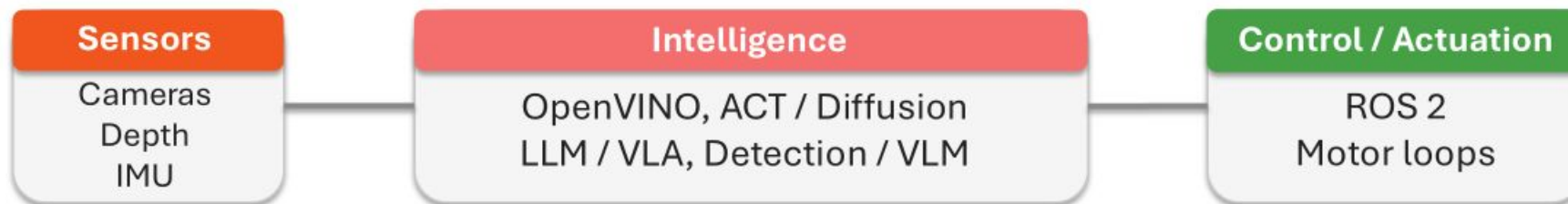


*Intel brings a humanoid software stack in partnership with RedHat
— not just silicon — helping developers realize optimized AI solutions and real-time execution.*



Unitree Demo Runtime Stack: Core Ultra 3 + RHEL

Intel accelerates workloads across the entire humanoid robotics pipeline



Panther Lake Workload Acceleration



GPU efficient algorithms + inference



NPU efficient perception + policy inference



Path planning + CPU efficient algorithms



ROS 2 executors + control loops



Background services + comms



Red Hat

Red Hat Enterprise Linux

PREEMPT_RT
Linux kernel

ROS 2

SELinux + bootc
Secure immutable runtime

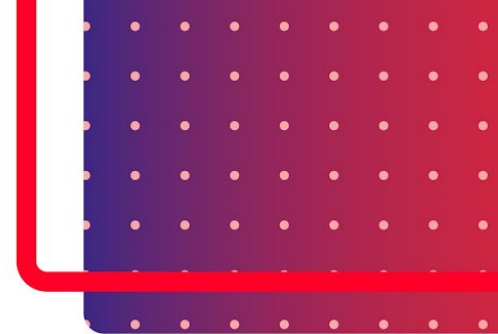
Fleet management

Humanoid demo story: Sense → Understand → Act on one SoC, with RHEL providing deterministic and secure runtime behavior



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Quick tip

Insert image in this designated area, deleting the shaded background. Keep the left, right, top, and bottom margins clear to maintain the open feel in accordance with the brand.



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- ▶ 1. Real-Time Kernel (PREEMPT_RT)
- ▶ The PREEMPT_RT kernel transforms RHEL into a deterministic powerhouse. It ensures critical motor control loops meet deadlines by minimizing latency, preventing the "jitter" that can destabilize hardware operations in industrial settings.
- ▶
- ▶ 2. Native ROS 2 Tier 2 Support
- ▶ RHEL is a Tier 2 platform for ROS 2 Jazzy Jalisco. This means core libraries are natively built and tested, offering a production-grade alternative to community distributions for enterprise-scale robotics deployment.





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